What Enables In-Context Learning in Transformers?

An Analysis of Attention Mechanisms

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Motivation

- Large language models can perform **in-context learning (ICL)** — solving tasks from examples in the prompt without updating parameters.
- Recent theory suggests transformers may perform **gradient-descent–like updates through attention**.
- Understanding this mechanism helps explain **how transformers generalize to new tasks**.

Research Question

How do different attention mechanisms affect a transformer's ability to perform in-context learning?

- We compare models that differ only in their attention mechanism.

Attention Mechanisms

Standard Dense Attention
- Baseline transformer attention

Linear Attention
- Attention mechanisms with computation linear in sequence length
- Enables more efficient scaling with longer contexts

Sparse Attention
- Restricts attention to a subset of tokens
- Reduces computational complexity

Gated Linear Attention
- Uses gating mechanisms to modulate attention contributions from past outputs

Low-Rank Attention
- Approximates attention using low-rank projections to reduce complexity

Grouped Query Attention (GQA)
- Shares key and value heads across groups of query heads
- Used in modern efficient large language models

Experimental Design

- Our **goal** is to **determine whether optimization-like in-context learning is a general property of attention mechanisms or a consequence of specific architectural choices.**
- By holding the task distribution fixed and varying only the attention mechanism, we isolate how architectural constraints affect a model's ability to perform in-context learning.

Method

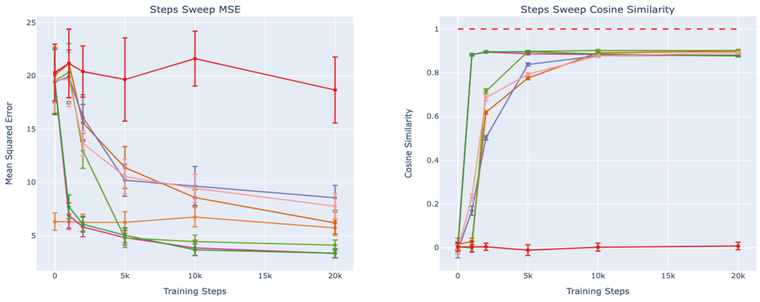
- To evaluate how attention mechanisms influence in-context learning, we measure both prediction performance and alignment with gradient descent.
- Prediction error is measured using **mean squared error (MSE)** on held-out regression tasks.
- To test whether the models behave like optimization procedures, we compute the **cosine similarity** between the model's implicit update direction and the analytic **one-step gradient descent update**.

Experiment Sweeps

T-step GD **GLA** **GQA** **Kernelized Linear**

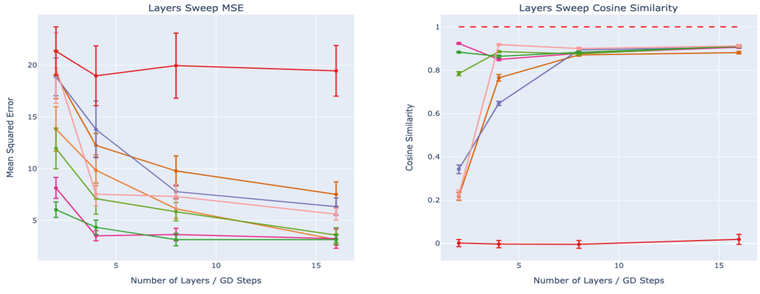
Low-rank k0.5 **LSA** **Softmax** **Sparse Causal**

1. **Training/Test Sweep**
   - Evaluate generalization across different training and testing configs



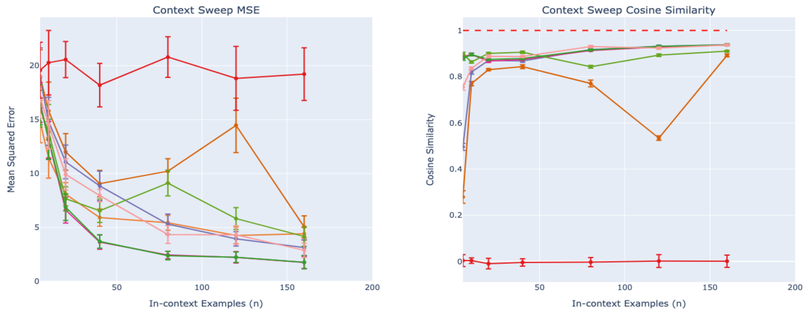
   - Most attention mechanisms improve with training, reducing prediction error and increasing alignment with gradient descent. Sparse attention fails to converge.

2. **Layer Sweep**
   - Analyse how increasing the number of attention layers affects ICL



   - Increasing transformer depth improves in-context learning, supporting the interpretation that layers correspond to iterative optimization steps.

3. **Context Sweep**
   - Measure performance as the number of examples provided in the prompt increases.



   - Performance improves with more in-context examples, indicating that transformers use contextual data to infer the underlying regression weights.

Results

- **Linear attention (LSA) mechanisms most closely emulate gradient descent, achieving both low prediction error and strong optimization alignment.**

Training Sweep
- Performance improves with training
- Error decreases and gradient alignment increases
- Linear attention converges fastest

Layer Sweep
- More transformer layers improve performance
- Deeper models show stronger gradient alignment
- Supports layers acting like iterative optimization steps

Context Sweep
- More in-context examples improve predictions
- Models use context to infer regression weights

All attention mechanisms except sparse attention demonstrated strong in-context learning performance, suggesting that efficient attention architectures can maintain performance without sacrificing learning capability

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